# Real Cosmology

A Proton–Electron Universe from Two Fluctuating Potentials:
a tensioned vacuum, variable-sign mass and charge, and a nuclear-powered bounce

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*Mathematical theory and formulation by the author; manuscript developed in collaboration with Claude (Anthropic). See Section 11.*

**Abstract**

We sketch an exploratory cosmology built from the RealQM / RealNucleus program, in which all structure is Coulombic (no strong force, no elementary neutron). The universe rests on a **tensioned elastic vacuum**: a continuum whose ground state has zero deflection but nonzero tension, the medium on which the Laplacian acts. On it live **two fluctuating potentials**, a gravitational $\varphi_g$ and an electric $\varphi_E$. Mass and charge are not postulated but read off as the curvature of the potentials, $\rho = \nabla^2\varphi$, which makes both *variable in sign* and *net-zero* ($\int \nabla^2\varphi\, dV = 0$): the universe carries no net mass and no net charge, the latter giving equal numbers of protons and electrons with no baryogenesis. The Laplacian amplifies small scales ($\rho_k \propto k^2\varphi_k$). A positive-mass region collapses gravitationally into a hot dense state; its matter separates into **equal-mass protons and electrons**; deuterons form, and the released binding energy drives a **nuclear-powered bounce** into expansion, while negative-mass regions repel as **dark energy**. The electron is taken to have **two temperature-selected size-phases** — compact (inside nuclei, ∼MeV) and expanded (atomic, ∼eV) — so a single cooling curve orders nuclei then atoms, with neutron = proton + compact electron, the weak interaction as the size-transition, neutron stars as forced compactification, and the primordial helium fraction set by a transition temperature $T_*$. We state the picture, its one quantitative handle ($T_*$), and its load-bearing open problems — chiefly *what sets the nuclear scale*, whether the *bounce energy budget* beats gravity, and whether the $\nabla^2\varphi$ *fluctuation spectrum* can match the CMB.

**Status.** This is an exploratory proposal, not established cosmology. It departs sharply from the Standard Model and ΛCDM and must eventually be reconciled with them; the open problems in Section 9 are load-bearing. Read it as "what follows *if*."

**Keywords:** cosmology; bounce cosmology; induced gravity; proton–electron model of the nucleus; dark energy; primordial nucleosynthesis; RealQM; multiphase Coulomb continuum mechanics.

## 1 Introduction

The RealQM program formulates quantum mechanics as multiphase 3D continuum mechanics — unit-charge densities on non-overlapping domains separated by Bernoulli free boundaries, minimising the Coulomb energy [1]. Its nuclear extension, RealNucleus, revives the pre-1932 proton–electron picture (neutron = proton + electron) and reproduces nuclear binding from pure Coulomb interactions with no strong-force postulate [2]. This note asks what cosmology results if the *same* ingredients — Coulomb attraction, a multiphase free-boundary vacuum, and particles built from charge — are taken as the starting point of the universe. The result is a collapse-and-bounce cosmology with dark energy and a unified treatment of nuclear and atomic structure built in. We develop it as a chain of postulates and consequences, and we keep the open problems explicit.

## 2 The substrate: a tensioned vacuum

A Laplacian needs something to differentiate. The operator $\nabla^2\varphi$ is the curvature-restoring force of an elastic medium: a taut string's restoring force is $T\,\partial^2 u/\partial x^2$ (tension times curvature), and the wave equation is $\mu\,\partial_t^2 u = T\,\nabla^2 u$. The construction below therefore presupposes a **pre-stressed elastic vacuum** — a continuum whose ground state has *zero deflection but nonzero tension.* The potentials are deflection fields of this substrate; mass and charge are its curvature. This is the hidden structure of every field theory (the gradient term $\frac{1}{2}(\nabla\varphi)^2$ in any Lagrangian *is* an elastic tension) and the spirit of Sakharov's induced gravity — gravity as the elasticity of the vacuum [4].

The tension is not optional: it sets the wave speed $c = \sqrt{T/\mu}$ (what lets light propagate), and it stores energy even at zero deflection — a string tightened from slack holds elastic energy. That standing vacuum energy is the cosmological-constant / dark-energy term, so the substrate tension and the negative-mass repulsion of Section 3 may be one and the same.

This substrate is the real light-carrying medium of *Many-Minds Relativity* [3]: $c = \sqrt{T/\mu}$ is its *wave speed* (a property of the medium, not an Einstein postulate). From the wave momentum $P = mc$ with $E = Pc$ one obtains $E = mc^2$ ([3], ch. 16) — mass–energy equivalence as a medium-wave result, with no Lorentz-transformation kinematics. The medium's tension also supplies the *Poincaré stresses* that hold each charge cloud together against its Coulomb self-repulsion (the Bernoulli free boundary of RealQM).

But *mass is not electromagnetic*, and the substrate does not do double duty as both. The electromagnetic sector carries *charge*; *mass is inertia* — the localization, i.e. the frequency / inverse size of the matter wave, the dispersion gap $\omega_0 = mc^2/\hbar$ — a separate, gravitational quantity (it is signed, hence cannot be the positive-definite EM field energy). Electromagnetism and gravitation meet *only through energy* ($E = mc^2$), embodied in matter (a charge welded to an inertia, the charge-to-mass ratio the exchange rate), not by sharing one substrate.

Where does the tension come from? Two routes: **(a) fundamental**, a brute constant of the substrate (like $c, G, \hbar$), the ground state simply taut; or **(b) self-tuned**, a slack symmetric state falling into a lower-energy tensioned state by spontaneous symmetry breaking (as water freezes, or the Higgs settles into its valley), so that energy minimisation, not an external agent, does the tuning. Route (b) still presupposes a substrate with the *capacity* for tension and laws that favour breaking. Physics can describe the string and how it self-tightens, but "why a tensioned something rather than nothing" is not an empirical question. We take the tensioned vacuum as the *irreducible given* of the model and state it plainly rather than smuggle it in.

## 3 Two fluctuating potentials: mass and charge as curvature

The fundamental objects are two potentials on the substrate of Section 2 — a gravitational $\varphi_g$ and an electric $\varphi_E$ — each carrying a small-scale fluctuation. Density is not assumed; it is the Laplacian of the potential (Poisson, run backwards):

$$\rho_{\text{mass}} = \frac{1}{4\pi G}\,\nabla^2\varphi_g, \qquad \rho_{\text{charge}} = -\varepsilon_0\,\nabla^2\varphi_E. \tag{1}$$

Three consequences follow at once.

1. **Variable sign.** $\rho$ is positive where $\varphi$ is concave and negative where convex. Gravity thus has $\pm$mass and electromagnetism $\pm$charge, both *derived* from one potential each, not postulated as separate ingredients.

2. **Net zero.** For any localised fluctuation, $\int \nabla^2\varphi\, dV$ is a surface term that vanishes, so $\int \rho\, dV = 0$. The universe carries **zero net mass and zero net charge** automatically. Charge neutrality means equal numbers of protons and electrons — $\#p = \#e$ — with *no baryogenesis required*; mass neutrality means the $\pm$mass regions balance.

3. **Small-scale amplification.** For a Fourier mode, $\rho_k \propto k^2\varphi_k$: the Laplacian amplifies short wavelengths by $k^2$, so a gentle potential ripple becomes a large density contrast. We refer to this as the "inflation" of the density field; it is amplification, not the exponential spatial expansion of standard inflation (Section 9).

Gravity acts on the large scale (same-sign mass attracts, opposite repels), separating matter into positive-mass regions and negative-mass voids; the electric potential structures the small scale into $\pm$charge appearing as protons and electrons — a neutral plasma.

# 4 The collapse–bounce and dark energy

The cosmic sequence runs as a self-regulating collapse–bounce:

1. A positive-mass region **collapses gravitationally**, converting potential energy into heat: the hot dense state is *generated*, not assumed.

2. In it, the electric potential splits the positive-mass matter into **equal-mass protons and electrons**. (Mass-sign and charge-sign are independent: these protons and electrons both carry positive mass.)

3. **Deuterons form and release energy**, and that energy drives expansion — reversing the collapse. A nuclear-powered bounce.

4. Expansion $\rightarrow$ cooling $\rightarrow$ atoms; the negative-mass regions sustain ongoing accelerated expansion as dark energy.

Two features stand out: the picture *avoids an initial singularity* (a collapse-and-bounce, not a Big Bang from nothing, in the spirit of bounce cosmologies [5]), and it *unifies the expansion drivers* — the initial expansion from the nuclear bounce, the ongoing acceleration from negative-mass repulsion. The load-bearing test is the energy budget: the released nuclear binding energy must exceed the *gravitational* binding of the collapsing region for the bounce to occur (Section 9).

# 5 Equal-mass charges and the two electron sizes

We take protons ($+1$) and electrons ($-1$) to be **equal in mass** at the level of the charge fluctuation. In the proton–electron picture a neutron is $p+e$, so any neutral atom $(A, Z)$ contains $A$ protons and $A$ electrons: $A-Z$ compact inside the nucleus (one per neutron) and $Z$ expanded in orbitals. Equal numbers of protons and electrons everywhere is consistent with the net-zero charge of Section 3.

The electron is taken to occupy **two size-phases**, its extent adapting to the Coulomb well: *compact* in a deep well (inside a nucleus, $\sim$fm, $\sim$MeV) and *expanded* in a shallow one (atomic, $\sim$Å, $\sim$eV). The same particle then spans the nuclear and atomic energy scales, a ratio $\sim 10^5$–$10^6$. Chemistry and nuclear physics become one Coulomb free-boundary mechanism at two scales: a molecule is nuclei glued by *expanded* (valence) electrons; a nucleus is protons glued by *compact* electrons. What physically sets the separation is the central open problem (Section 9).

**Why the gap is $\alpha^2$.** The electron has three characteristic lengths — the three ways to build a length from $(\hbar, m, c, e)$, each differing by one factor of $\alpha = e^2/\hbar c$:

$$\underbrace{a_0 = \frac{\hbar^2}{me^2} \approx 0.5\,\text{Å}}_{\text{atomic }(\hbar,e)} \xrightarrow{\times\alpha} \underbrace{\bar{\lambda}_C = \frac{\hbar}{mc} \approx 386\,\text{fm}}_{\text{Compton }(\hbar,c)} \xrightarrow{\times\alpha} \underbrace{r_e = \frac{e^2}{mc^2} \approx 2.8\,\text{fm}}_{\text{classical / nuclear }(e,c)}, \qquad a_0 : \bar{\lambda}_C : r_e = 1 : \alpha : \alpha^2. \tag{2}$$

The two electron *phases* are the ends — expanded (atomic, $a_0$; non-relativistic Coulomb) and compact (nuclear, $r_e$; electromagnetic mass). The middle length, the *Compton wavelength* ($\hbar, c$; the quantum-relativistic scale where confinement costs $\sim mc^2$), is the quantum rung between them, which is why the nuclear/atomic gap is $\alpha^2$ (two steps of $\alpha$). Dropping $\hbar$ carries one from the atomic to the classical scale; $\alpha$ is the price of each step. (The value of $\alpha$ itself is an input, Section 9.)

**The two sizes are a Bernoulli effect, set without mass.** Described by charge and a kinetic coefficient $\kappa$ alone (the electrostatic structure needs no mass), the electron takes two forms: an *expanded* shell whose density tapers to zero (the H atom — a vanishing outer boundary forces a large, gently-varying cloud), and a *compact* core *caged* between protons whose density stays finite at the Bernoulli free boundary (no steep-taper penalty, so a small cloud is allowed). Minimising $\kappa \cdot$ (kinetic) + Coulomb gives $L = 2\kappa b/Ze^2$, and the size ratio $L_{\text{small}}/L_{\text{big}} = (b_2/b_1)/Z_{\text{eff}}$ is *independent of* $\kappa$ — pure geometry and boundary condition. If the caged electron reaches the classical scale, the ratio is the mass-free number $r_e/a_0 = \alpha^2 \approx 1/18800$ (the deuteron relation); non-relativistic Coulomb caging gives only $\sim 1/Z_{\text{eff}}$. Mass enters only as inverse size: the energy is the square-root dispersion $E = \sqrt{(\hbar c/L)^2 + (mc^2)^2}$, the rest mass the gap ($L \to \infty$) and the size the momentum term — so mass is the matter-wave's inverse Compton wavelength (the dispersion gap), the *gravitational* charge, distinct from electric charge.

**Empirical check.** Observed sizes: the H atom $\approx a_0 = 0.53$ Å $\approx 53{,}000$ fm; the deuteron nucleus $\approx 2.1$ fm (rms). Their ratio $\approx 4 \times 10^{-5} \approx 1/25{,}000$ matches $\alpha^2 \approx 5 \times 10^{-5} \approx 1/18{,}800$ to within $\sim 30\%$, with the deuteron ($\sim 2.1$ fm) sitting essentially at the classical electron radius ($r_e = \alpha^2 a_0 = 2.8$ fm). So the $\alpha^2$ nuclear/atomic size relation is confirmed to order of magnitude — the H-atom/deuteron-nucleus size ratio comes out as $1/\alpha^2$, from charge, $\hbar$ and $c$ alone, with no strong force. (The deuteron is unusually loosely bound, so this is the scale/ratio being right, not a precise fit.)

# 6 Temperature, the cosmic sequence, and the weak interaction

A bound state survives once the bath can no longer ionise it ($kT <$ binding). The two electron phases therefore switch on at two temperatures set by their two binding energies, forcing the order plasma $\to$ nuclei $\to$ atoms (Table 1). The temperature ratio between the two epochs ($\sim 10^6$) equals the energy-scale ratio — one fact, not two.

| epoch | process | temperature |
|---|---|---|
| plasma | free $\pm$ charges | $> 10^{10}$ K |
| nuclear condensation | compact $e$ captured $\to$ nuclei | $\sim 10^{10}$ K ($kT \sim$MeV) |
| recombination | expanded $e$ captured $\to$ atoms; CMB | $\sim 3000$ K ($kT \sim$eV) |

Table 1: The two condensation epochs, separated by $\sim 10^6$ in temperature.

The compact electron is precisely the electron inside a neutron: neutron $= p +$ compact $e$. The two weak processes are then the two directions of the size-transition, driven by temperature/density: *compactify* ($p + e \to n$, electron capture) and *expand* ($n \to p + e$, $\beta$-decay). This lands on real astrophysics: crushing matter forces electrons compact $\to$ wholesale $p \to n \to$ a neutron star (neutronisation in core collapse). The cosmic neutron/proton ratio at freeze-out is the compact/free electron ratio at $T_*$.

## 7 The quantitative handle: $T_*$ and the helium fraction

The postulate predicts a transition temperature $T_*$ at the compactification energy. The fraction of electrons that compactify by freeze-out fixes the cosmic neutron/proton ratio and hence the **primordial helium fraction**; matching the observed $\sim$25% He ($n/p \approx 1/7$) is the decisive quantitative test. In standard physics freeze-out is at $kT \approx 0.8$ MeV ($T \approx 10^{10}$ K), tied to the $n$–$p$ mass difference 1.29 MeV, so the prediction is $T_* \sim$ MeV $\sim 10^{10}$ K. Equivalently, the universe's small:big (compact:expanded) electron ratio equals the $N$:$Z$ of primordial matter, $\approx$1:7, dominated by hydrogen (which has no compact electron).

A neutron is a proton with a compact electron, so the neutron/proton ratio is the compact/free electron ratio at freeze-out, $(n/p)_f = e^{-E_*/kT_f}$, with the compactification gap $E_*$ playing the role of the $n$–$p$ mass difference; free neutrons partly expand back ($\beta$-decay, factor $d$) before nucleosynthesis, and essentially all survivors end in He-4:

$$(n/p)_{\rm nuc} = \frac{(n/p)_f\, d}{1 + (n/p)_f\,(1-d)}, \qquad Y = \frac{2\,(n/p)_{\rm nuc}}{1 + (n/p)_{\rm nuc}}. \tag{3}$$

With the model's single MeV-scale handle ($E_* \approx 1.29$ MeV, $T_f \approx 0.8$ MeV so $T_* \sim 10^{10}$ K, $d \approx 0.74$): $(n/p)_f = 0.199 \to (n/p)_{\rm nuc} = 0.140 \to \boxed{Y = 0.245}$, the observed primordial helium. A single compactification energy thus reproduces the $\sim$25% helium — the model's *first falsifiable quantitative success*. (Caveat: $T_f$ is, in standard BBN, derived from the weak-rate-vs-expansion balance; here it is a second input, so the genuine prediction is the ratio $E_*/kT_f \approx 1.6$ with $E_* \sim 1.3$ MeV.)

## 8 Energetics: where the binding energy goes

Forming a deuteron releases its binding ($\sim$MeV). Because the binding is Coulombic — an electron dropping from the expanded into the compact phase — the energy is released as photons, exactly as an atomic electron radiates when it falls to a lower level, scaled up $\sim 10^5$ to $\sim$MeV gammas. Equivalently it appears as the deuteron's mass defect ($E = mc^2$): the deuteron is lighter than its constituents by binding/$c^2$. Cosmologically it is dumped into a photon bath with $\sim 10^9$ photons per baryon, a $\sim 10^{-8}$ perturbation that redshifts to insignificance — the universe is not noticeably reheated.

The process is **not self-sustaining**: the released gammas feed the same bath that re-dissociates fresh deuterons while $kT \gtrsim$ binding, so net formation proceeds only as the universe *cools* below $T_*$ (the deuterium bottleneck). This is the opposite of barrier-limited stellar fusion (favoured by heating, energy release sustaining the high $T$ it needs); deuteron *formation* is exothermic capture (favoured by cooling) and self-quenches at high $T$. Proton density sets the *yields* (as the baryon density does in standard nucleosynthesis), not ignition. The one self-sustaining channel is *gravitational*: in a collapsing dense region, rising density forces electron capture in a runaway sustained by gravity — neutron-star neutronisation.

## 9 Open problems

1. **The nuclear scale.** Non-relativistic equal-mass Coulomb is scale-free (a compact $\sim$fm electron would need glue $\sim 5 \times 10^4\, m_e$). *Candidate resolution:* the equal-mass cloud sits at its *classical radius* $r = e^2/m_e c^2 = \alpha^2 a_0 \approx 2.8$ fm (Coulomb self-energy = rest energy),

energy $\sim m_e c^2 \approx 0.5$ MeV, so the nuclear/atomic gap is $1/\alpha^2 \approx 1.9 \times 10^4$ — derived from the fine-structure constant, not the strong force. The compact electron is reached by *caging* (trapped between protons, $H_2^+$-like, compressed), the expanded one by *attraction* (one proton, relaxing to $a_0$). The step it needs is the *electromagnetic-mass* / field-self-energy relation ($P = mc \Rightarrow E = mc^2$, in the sense of Many-Minds Relativity [3], *not* Einstein SR), which the non-relativistic engine lacks — why it yields binding *ratios* but not the absolute MeV.

2. **The bounce energy budget.** Does the released nuclear binding energy exceed the gravitational binding of the collapsing region, so the bounce of Section 4 actually reverses collapse? At large scales gravity typically dominates nuclear energy (why supernova cores collapse rather than explode on fusion alone). The inequality $E_{\text{nuclear}} \gtrsim E_{\text{grav}}$ at the relevant scale must be checked.

3. **The fluctuation spectrum and $\varphi$ dynamics.** $\rho = \nabla^2\varphi$ gives a blue ($k^2$-amplified) density spectrum, whereas the CMB and large-scale structure fit a near scale-invariant one ($n_s \approx 0.96$); the potential must carry a compensating red spectrum. Moreover $\rho = \nabla^2\varphi$ is a definition, not yet a law of motion — $\varphi$ needs its own field equation.

4. **Equal mass vs. $m_p/m_e = 1836$.** Here $\alpha$ and 1836 are *independent axes*: $\alpha$ fixes the nuclear/atomic *scale* gap (the classical radius above), while 1836 fixes the proton/electron *inertia* asymmetry; the two do not reduce to each other. Equal mass buys the scale result and defers 1836, the open input — which is *not derived in the Standard Model* either (proton mass $\sim$99% QCD binding energy, electron mass from the Higgs). Candidate direction: the proton as a relativistically compacted electron (energy $\sim$938 MeV at the proton Compton wavelength, so 1836 is the compaction ratio — suggestive but circular). Numerological aside: $m_p/m_e \approx 6\pi^5 = 1836.1$.

5. **The neutrino.** $\beta$-decay has a continuous spectrum — the 1930 objection that sank the proton–electron neutron and forced Pauli's neutrino [6]. The "electron expands" step needs a home for that energy and spin.

6. **Precision data and the rest.** BBN abundances and the CMB acoustic peaks are percent-level successes the sequence must reproduce; antimatter and proton substructure must be accounted for in a two-ingredient ontology.

## 10 Computational status

What the RealQM/RealNucleus engine has shown, versus what remains. **Done:** *p*-*e*-*p* binds as a stable configuration; the He-4/deuteron binding *ratio* comes out 13.1 vs. experiment 12.7 [2]; the lepton-mass scaling law $R \propto 1/m_{\text{glue}}$ is reproduced, with muon-catalysed fusion (the muonic molecular ion $dd\mu$ at $\sim$512 fm) as the observed proof that a heavy negative glue binds two protons; and the helium-fraction calculation of Section 7 gives $Y = 0.245$ from a single MeV-scale handle, matching the observed primordial helium. **Open:** the two-scale test (does one engine give MeV *and* eV in the same units? equal-mass Coulomb returns $\sim$1, confirming a scale ingredient is needed); and a cooling-quench "mini Big Bang" using the engine's temperature and Brownian-dynamics machinery to enact the two condensation epochs.

## 11 Collaboration

The mathematical theory and the cosmological proposal are the author's. The manuscript was developed in dialogue with Claude (Anthropic), which also assisted in articulating the argument, surfacing the quantitative constraints and open problems, and preparing the supporting interactive simulations of the RealQM/RealNucleus program. We present the human–AI collaboration transparently as part of the working method.

**One-sentence summary.** If protons and electrons are equal-mass $\pm$ charges read off as the curvature of two fluctuating potentials on a tensioned vacuum, and the electron has two temperature-selected sizes, then nuclei, atoms, the weak interaction, neutron stars, dark energy, and the primordial helium fraction follow from one cooling curve — a unifying picture whose survival hinges on a single hard question: what sets the nuclear scale, if not the strong force.

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